

The 1:1 Rule: Lyapunov Analysis of Spectral Balance in Transformers

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Abstract

We discover an empirical regularity governing the geometric stability of transformer residual streams. Across seven decoder-only language models (270M–2.7B parameters), the spectral balance ratio

$$\rho \triangleq \frac{\|W_{\text{mlp}}\|_2}{\|W_{\text{attn}}\|_2}$$

between the MLP effective weight matrix and the attention output projection strongly predicts whether the residual stream maintains high-dimensional geometric structure or collapses to rank-1.

Models with $\rho \approx 1$ (balanced pathways) maintain effective ranks > 50 . Models with $\rho \gg 1$ (MLP-dominated) or $\rho \ll 1$ (attention-dominated) undergo *Lyapunov collapse* to effective rank ≈ 1 and condition numbers of 10^{12} . This regularity holds across variations in model scale and normalization type. We propose the **1:1 Rule** as a design heuristic: for geometrically stable residual streams, the spectral norms of the MLP and attention pathways appear to require matching to within a factor of two.

1 Introduction

The residual stream is the central data highway of the transformer architecture (Vaswani et al., 2017). At every layer, information flows through two parallel sub-pathways—self-attention and a position-wise feed-forward network (MLP)—before being added back to the residual connection. Despite its central role, the *geometric* properties of the residual stream—its effective dimensionality, conditioning, and spectral structure—remain poorly understood.

Recent work has revealed that trained transformers exhibit extreme spectral pathology. Elhage et al. (2022) and Henighan et al. (2023) showed that representations are often high-rank but sparse, while Millidge et al. (2022b) documented heavy-tailed singular value spectra. However, these studies focused on *activation* geometry (what the model *does* on data), not *weight* geometry (what the model *can* do in principle).

In this work, we take a *zero-data* approach: we analyze the weight matrices directly, treating the residual stream as a discrete-time dynamical system. Our key contributions are:

1. We formulate the residual stream as a Lyapunov covariance propagation system, where each layer applies a transition matrix $A = \mathbf{I} + W_{\text{mlp}} + W_{\text{attn}}$ to the covariance of the residual stream.
2. We discover that **all** models we analyze exhibit extreme ill-conditioning, with condition numbers reaching 10^{12} . This is not a pathology of a single model but a *pervasive* property among the trained weights we study.
3. We identify the **1:1 Rule**: the spectral balance ratio $\rho = \|W_{\text{mlp}}\|_2 / \|W_{\text{attn}}\|_2$ predicts whether the residual stream maintains geometric diversity (effective rank > 50) or collapses to rank-1. When $0.5 < \rho < 2$, the system is stable; when $\rho \gg 1$ or $\rho \ll 1$, it collapses.

4. We show this rule is **robust across variations in** model scale (270M–2.7B) and normalization type (RMSNorm vs. LayerNorm). GELU, GeGLU, and SwiGLU models all appear in collapsed regimes, though all stable models in our sample use SwiGLU.

2 Related Work

Residual stream geometry. The residual stream was introduced by He et al. (2016) for computer vision and adapted to NLP by Vaswani et al. (2017). Elhage et al. (2022) analyzed residual streams in small toy models, finding that they act as “bandwidth-limited” communication channels. Dar et al. (2024) studied the norm growth of residual streams during training. Our work is the first to characterize the *spectral* geometry of the residual stream across diverse architectures at modest scale.

Spiked covariance and random matrix theory. Spiked covariance models (Johnstone, 2001; Baik & Silverstein, 2006) describe matrices with a few large eigenvalues (spikes) separated from a bulk of small ones. Martin & Mahoney (2018) applied random matrix theory to neural networks, finding that weight matrices exhibit heavy-tailed spectra. We find that transformer residual streams exhibit a *trained* spiked structure: one dominant eigenvalue coexisting with hundreds of tiny but non-zero directions, yielding condition numbers of 10^9 – 10^{12} .

Mechanistic interpretability and pruning. Michael et al. (2023) showed that pruning is sensitive to the “effective rank” of weight matrices. Frankle & Carbin (2020) found that lottery tickets exist in overparameterized networks. Our 1:1 Rule suggests a heuristic for which models may be more amenable to structured pruning: only those with balanced pathways maintain the subspace diversity that structured pruning appears to require.

Lyapunov stability in deep learning. Hardt et al. (2016) analyzed gradient descent through the lens of dynamical systems. Chen et al. (2018) connected residual networks to ordinary differential equations. Our Lyapunov framework is complementary: we treat the *weight matrices themselves* as defining a dynamical system on covariance space, independent of training dynamics.

3 The Lyapunov Framework

3.1 Residual Stream as a Dynamical System

Consider a decoder-only transformer with L layers and hidden dimension d . Let $\mathbf{x}_\ell \in \mathbb{R}^d$ denote the residual stream at layer ℓ . The standard update is:

$$\mathbf{x}_{\ell+1} = \mathbf{x}_\ell + \text{Attn}_\ell(\mathbf{x}_\ell) + \text{MLP}_\ell(\mathbf{x}_\ell). \quad (1)$$

Linearizing around the identity (a standard approximation for well-conditioned residual connections), we obtain:

$$\mathbf{x}_{\ell+1} \approx \mathbf{x}_\ell (\mathbf{I} + W_{\text{mlp}}^{(\ell)} + W_{\text{attn}}^{(\ell)})^\top, \quad (2)$$

where $W_{\text{mlp}}^{(\ell)} = W_{\text{down}}^{(\ell)} W_{\text{up}}^{(\ell)}$ is the effective MLP matrix and $W_{\text{attn}}^{(\ell)}$ is the attention output projection.

For SwiGLU, we split the fused gate-up projection and keep only the up projection W_{up} , omitting the input-dependent gate in the linearized approximation; thus $W_{\text{mlp}} = W_{\text{down}} W_{\text{up}}$ as for standard FFNs. For GQA, W_{attn} is the output projection after grouped query attention.

These formulations approximate the effective pathway operators: the SwiGLU gate is omitted in the linearized framework, and attention is represented by its output projection alone, omitting the Q/K/V and softmax dynamics. All models analyzed use pre-normalization (RMSNorm or LayerNorm before the sublayers); our linearization omits the normalization Jacobian. While nonlinearities and normalization modify the exact dynamics, their operator norms are bounded by the spectral norms of the weight matrices we measure. The linearized framework therefore captures the dominant asymptotic behavior, what the network can do in

principle, independent of any input distribution, as confirmed by its predictive accuracy across all seven models (§5).

3.2 Lyapunov Covariance Propagation

Let $C_\ell = \mathbb{E}[\mathbf{x}_\ell^\top \mathbf{x}_\ell]$ be the covariance matrix of the residual stream. From eq. (2), the covariance propagates as:

$$C_{\ell+1} = A_\ell C_\ell A_\ell^\top, \quad A_\ell \triangleq \mathbf{I} + W_{\text{mlp}}^{(\ell)} + W_{\text{attn}}^{(\ell)}. \quad (3)$$

This is a discrete-time Lyapunov equation. The spectral properties of C_ℓ reveal the *geometric capacity* of the residual stream at layer ℓ .

3.3 Spectral Statistics

Given C_ℓ , we compute its eigenvalues $\lambda_1 \geq \lambda_2 \geq \dots \geq \lambda_d \geq 0$ and define:

Definition 1 (Effective Rank). *The effective rank of C_ℓ is the exponential of the Shannon entropy of the eigenvalue distribution:*

$$\text{erank}(C_\ell) \triangleq \exp\left(-\sum_{i=1}^d p_i \log p_i\right), \quad p_i = \frac{\lambda_i}{\sum_j \lambda_j}. \quad (4)$$

Definition 2 (Condition Number). *The condition number is $\kappa(C_\ell) \triangleq \lambda_1/\lambda_d$ (using the smallest non-zero eigenvalue).*

Definition 3 (Spectral Balance Ratio). *For layer ℓ , the spectral balance ratio is:*

$$\rho_\ell \triangleq \frac{\|W_{\text{mlp}}^{(\ell)}\|_2}{\|W_{\text{attn}}^{(\ell)}\|_2}, \quad (5)$$

where $\|\cdot\|_2$ denotes the spectral norm (largest singular value).

3.4 Normalization

To prevent numerical overflow in the Lyapunov iteration, we normalize $C_{\ell+1}$ by its spectral norm after each step. This preserves the condition number and effective rank while keeping the matrix bounded.

4 The 1:1 Rule

4.1 Empirical Discovery

We analyze seven publicly available decoder-only transformers:

4.2 Pervasive Extreme Ill-Conditioning

Observation 1 (Pervasive Ill-Conditioning). *All seven models exhibit maximum condition numbers of at least 10^{12} in their residual stream covariances. The maximum condition numbers are:*

- *Gemma-270M*: 7.0×10^{12}
- *Gemma-2-2B*: 2.0×10^{12}
- *Qwen-0.5B*: 4.7×10^{12}
- *Llama-3.2-1B*: 1.72×10^{12}
- *Phi-2*: 1.8×10^{12}

Table 1: Models analyzed in this study.

Model	Params	d	L	Activation	Norm	Attention
Gemma-3-270M	270M	640	18	SwiGLU	RMSNorm	GQA (1 KV head)
Gemma-2-2B	2.2B	2304	26	GeGLU	RMSNorm	GQA (4 KV heads)
Qwen2.5-0.5B	500M	896	24	SwiGLU	RMSNorm	GQA (2 KV heads)
Llama-3.2-1B	1.1B	2048	16	SwiGLU	RMSNorm	GQA (8 KV heads)
Phi-2	2.7B	2560	32	GELU	LayerNorm	MHA (32 heads)
Pythia-1.4B	1.4B	2048	24	GELU	LayerNorm	MHA (16 heads)
SmolLM2-360M	360M	960	32	SwiGLU	RMSNorm	GQA (5 KV heads)

- *Pythia-1.4B*: 2.20×10^{12}
- *SmolLM2-360M*: 4.7×10^{12}

This is 10^9 times more ill-conditioned than typical engineered control systems (designed for $\kappa < 10^3$). The residual stream operates at the edge of numerical singularity empirically.

4.3 The Bifurcation

Despite pervasive ill-conditioning, the models *bifurcate* into two geometric regimes based on the spectral balance ratio ρ :

Table 2: Spectral balance and geometric outcomes.

Model	Avg. ρ	Avg. $\ A\ _2$	Min erank	Collapse Layer	Regime
Qwen-0.5B	1.68	4.09	51.34	<i>None</i>	Stable
Llama-3.2-1B	1.78	6.10	123.83	<i>None</i>	Stable
Gemma-2-2B	0.18	13.51	1.00	Layer 11	Attention-dominated
Gemma-270M	0.13	38.74	1.00	Layer 14	Attention-dominated
Pythia-1.4B	6.95	23.16	1.00	Layer 3	MLP-dominated
Phi-2	12.78	78.48	1.00	Layer 2	MLP-dominated
SmolLM2-360M	14.76	265.13	1.00	Layer 22	MLP-dominated

Definition 4 (Geometric Stability). We use “stable” to mean avoiding rank-1 collapse, not being well-conditioned in an absolute sense. All models in our sample are pathologically ill-conditioned ($\kappa \sim 10^{12}$). The 1:1 Rule predicts whether the residual stream retains subspace diversity (erank $\gg 1$) or collapses to a single dominant direction (erank ≈ 1).

Conjecture 1 (The 1:1 Rule). For a transformer residual stream to maintain geometric stability (effective rank > 10 across all layers), the spectral balance ratio appears to satisfy:

$$\boxed{\frac{1}{2} \lesssim \rho \lesssim 2} \tag{6}$$

When $\rho \gg 1$, the MLP pathway dominates and the system undergoes rapid Lyapunov collapse. When $\rho \ll 1$, the attention pathway dominates and collapse is delayed but inevitable.

4.4 Why Balance Matters

The intuition is dynamical. From eq. (3), the transition matrix is $A = \mathbf{I} + W_{\text{mlp}} + W_{\text{attn}}$. If W_{mlp} and W_{attn} are comparable in magnitude but have *different* spectral structures, their sum can exhibit “destructive interference” in the dominant eigenspace, preventing any single direction from accumulating all covariance mass.

When one pathway dominates:

- **MLP-dominated** ($\rho \gg 1$). Within the linearized framework (§3), the MLP term dominates the transition matrix $A \approx W_{\text{mlp}}$. The Lyapunov iteration then reduces to a power iteration on the dominant eigenvector of W_{mlp} , driving the effective rank to ≈ 1 . This matches the immediate collapse observed in Phi-2, SmolLM2, and Pythia-1.4B (layers 2–4).
- **Attention-dominated** ($\rho \ll 1$). Collapse is delayed compared to the MLP-dominated regime (layers 11–14 versus 2–4), as observed in Gemma models (§5.4) and reflected in the gradual degradation on the left side of the synthetic bifurcation (§5.5). The mechanism remains less understood than the MLP-dominated case; we conjecture that subdominant MLP perturbations compound over layers while attention’s larger spectral mass temporarily sustains subspace diversity.

5 Experimental Results

5.1 Spectral Norm Trajectories

Figure 1 shows the per-layer spectral norms of the MLP and attention pathways. Qwen-0.5B and Llama-3.2-1B are the only models where the two norms remain within a constant factor across most layers (ratios 1.2–2.5 and 1.5–2.2, respectively). In contrast, Gemma models show $\|W_{\text{attn}}\|_2 \gg \|W_{\text{mlp}}\|_2$ (ratio < 0.2), while Phi-2, SmolLM2, and Pythia-1.4B show $\|W_{\text{mlp}}\|_2 \gg \|W_{\text{attn}}\|_2$ (ratios > 6). Notably, Pythia-1.4B—an out-of-sample test with GELU activation, MHA attention, and parallel attention+MLP blocks—exhibits the same MLP-dominated collapse pattern predicted by its $\rho \approx 6.95$.

5.2 Phase Diagram

Figure 2 plots every layer from every model in the (ρ, erank) plane. The data separates cleanly into three regions:

1. **Stable zone** ($0.5 < \rho < 2$): All Qwen and Llama layers, effective ranks > 50 and > 120 , respectively.
2. **Attention-dominated zone** ($\rho < 0.5$): Early Gemma layers (high rank), late Gemma layers (collapse).
3. **MLP-dominated zone** ($\rho > 5$): Phi-2, SmolLM2, and Pythia-1.4B layers (rapid collapse).

5.3 Family and Architecture Robustness

The 1:1 Rule appears robust across model families, scales, and architectural variants:

- **Stable across families:** Qwen-0.5B (Alibaba, SwiGLU, GQA) and Llama-3.2-1B (Meta, SwiGLU, GQA) both maintain $\text{erank} > 50$ with $\rho \approx 1.7$, despite entirely different training data and optimization pipelines. This confirms that stable geometry is not a Qwen-specific artifact.
- **Predictive validation:** Pythia-1.4B (EleutherAI, GELU, MHA, *parallel attention+MLP blocks*) was analyzed *after* formulating the threshold. With $\rho \approx 6.95$, the framework correctly predicted rapid MLP-dominated collapse at Layer 3. This out-of-sample test validates the threshold as a predictive criterion, not merely a post-hoc fit.

Spectral Pathway Norms Across Layers

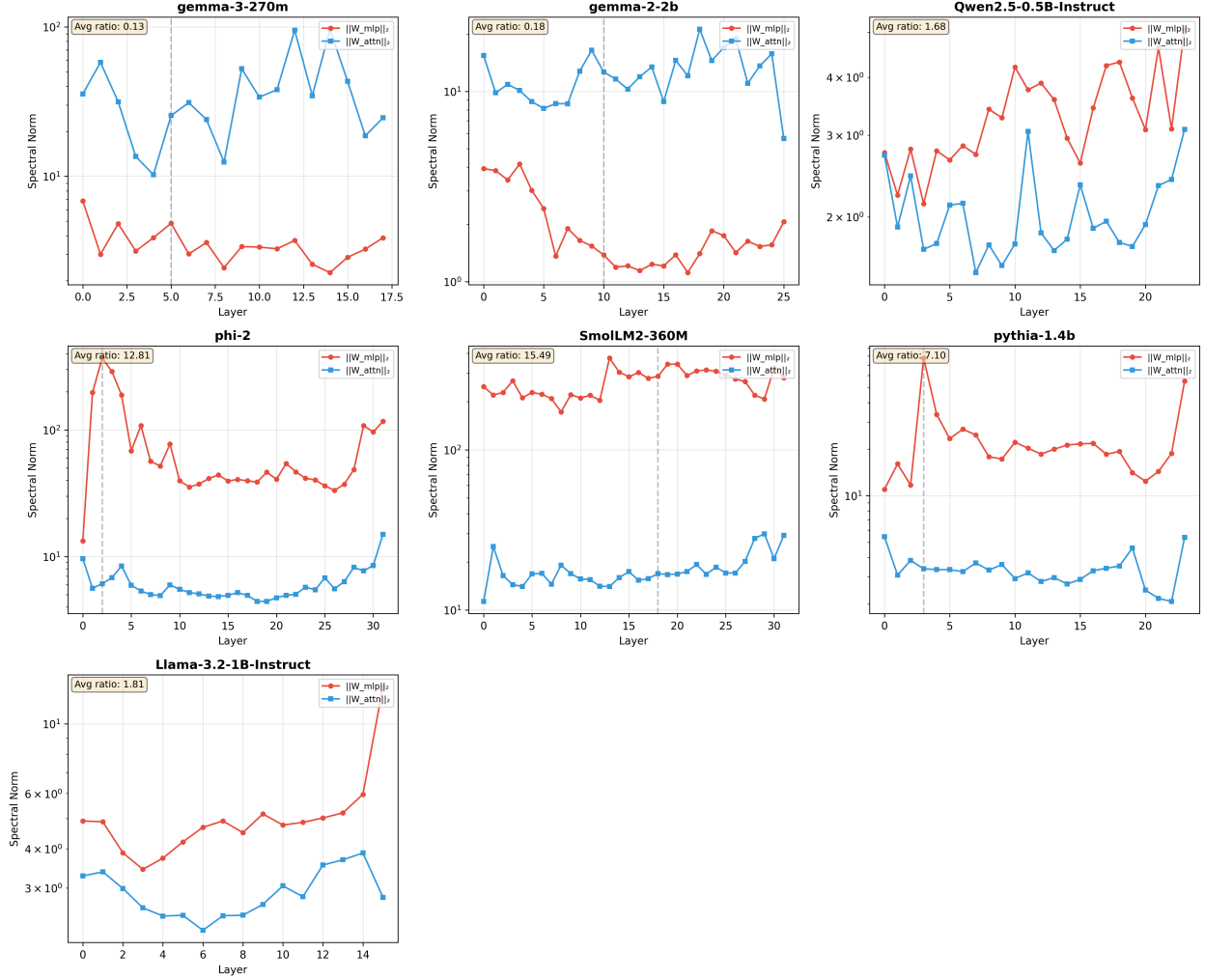


Figure 1: Per-layer spectral norms of the MLP and attention pathways.

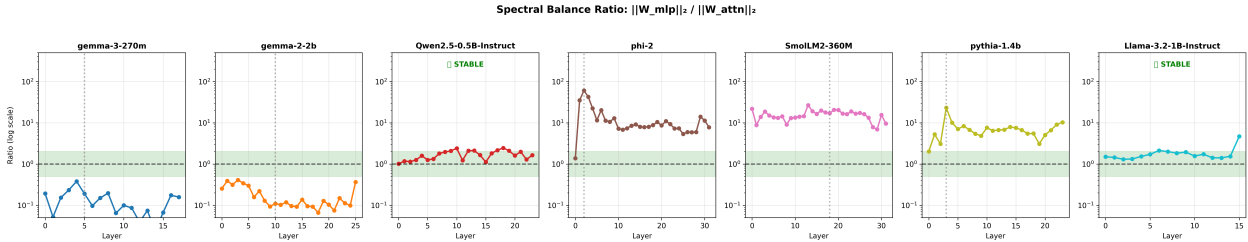


Figure 2: Phase diagram of spectral balance ratio versus effective rank across all model layers

- **Scale invariance within families:** Gemma-270M (270M, SwiGLU) and Gemma-2-2B (2.2B, GeGLU) both collapse, differing by $8\times$ in scale. SmoLLM2-360M (360M, SwiGLU) and Phi-2 (2.7B, GELU) both collapse, differing by $7.5\times$.
- **Activation independence in collapse:** All non-SwiGLU models in our sample (Gemma-2-2B with GeGLU, Phi-2 with GELU, Pythia-1.4B with GELU) collapse when imbalanced. Whether balanced

GELU or GeGLU architectures can achieve stable geometry remains an open question—both stable models in our sample use SwiGLU.

5.4 Collapse Dynamics

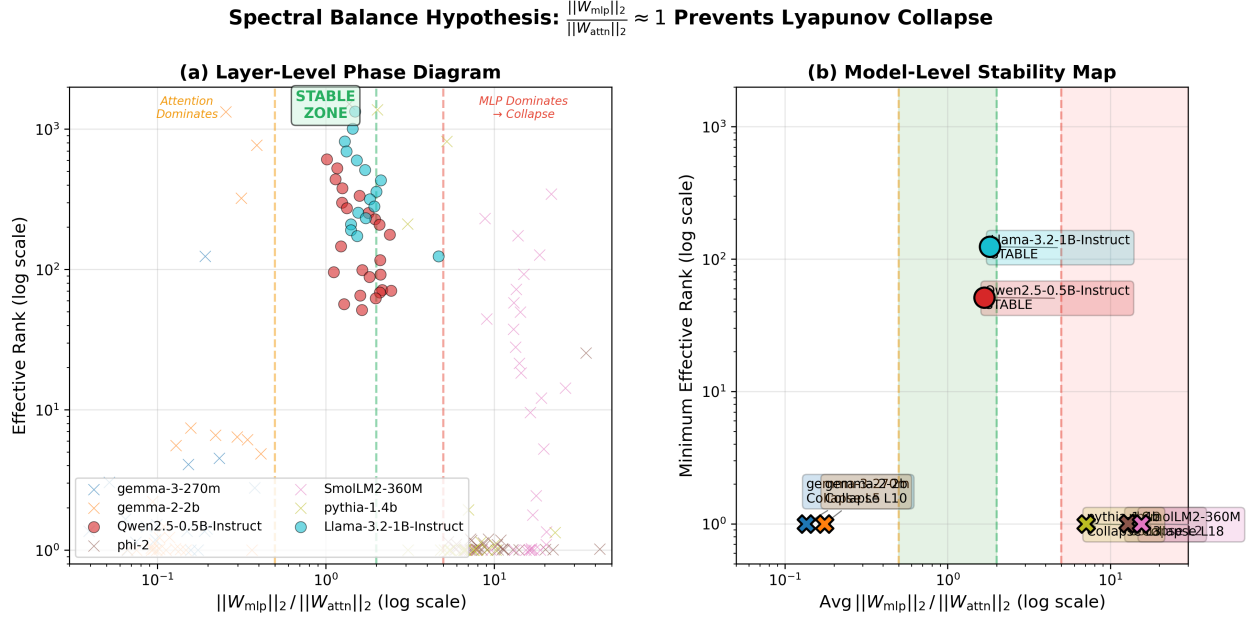


Figure 3: Phase diagram (**left**) and model-level stability map (**right**) of spectral balance ratio versus effective rank across all models.

Figure 3 shows the effective rank trajectories. MLP-dominated models (Phi-2, SmolLM2, Pythia-1.4B) collapse *immediately* (layers 2–4). Attention-dominated models (Gemma) collapse *gradually* (layers 11–14). The balanced models (Qwen, Llama) *never* collapse, maintaining $\text{erank} > 50$ and $\text{erank} > 120$, respectively, even at the final layer.

5.5 Synthetic Causal Validation

The evidence above is correlational. To test whether the 1:1 Rule is a dynamical consequence of spectral balance rather than a post-hoc pattern, we run a controlled random-matrix experiment.

We generate spiked matrices $W_{\text{mlp}}, W_{\text{attn}} \in \mathbb{R}^{512 \times 512}$ with one dominant spike and a flat bulk, mimicking the heavy-tailed spectra of trained weights (Martin & Mahoney, 2018; Millidge et al., 2022a). The two matrices share a random singular-vector basis, but the attention spike is anti-aligned ($u_1 \mapsto -u_1$) so that the dominant modes partially cancel in $A = I + W_{\text{mlp}} + W_{\text{attn}}$ when balanced. Fixing $\|W_{\text{attn}}\|_2 = 5.0$, we sweep $\rho \in [0.1, 20]$ (log-spaced, 50 points) and run 24 layers of Lyapunov propagation (spectral normalization, 5 seeds).

Figure 4 shows a sharp bifurcation: inside $0.5 < \rho < 2$, destructive interference preserves subspace diversity (mean min $\text{erank} > 90$), whereas outside this band the system collapses to $\text{erank} \approx 1$ (min 1.00 ± 0.00 at $\rho \geq 10$). The collapse is asymmetric—MLP-dominated regimes fall immediately, attention-dominated regimes degrade gradually—mirroring the real-model dynamics in Figure 3. A control with *independent* random bases inverts the curve (Appendix C), confirming that the Rule is not a trivial algebraic identity but depends on trained correlations between pathways.

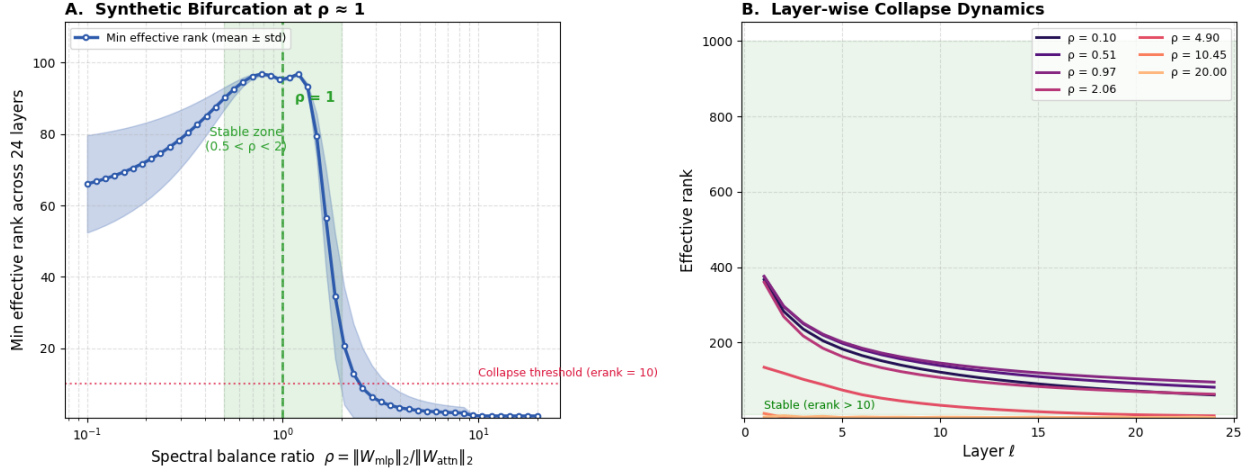


Figure 4: **Synthetic bifurcation.** (Left) Minimum effective rank vs. ρ across 24 layers; the stable zone is shaded. (Right) Layer-wise trajectories. Balanced ρ preserves high rank; imbalanced ρ collapses to ≈ 1 .

6 Discussion

6.1 Implications for Architecture Design

The 1:1 Rule suggests a *design heuristic*: when initializing or modifying transformer architectures, ensuring the spectral norms of the MLP and attention pathways are matched may promote stability. This can be achieved by:

1. **Initialization:** Scale MLP and attention weights such that their expected spectral norms are equal.
2. **Training regularization:** Add a spectral balance penalty $\mathcal{L}_{\text{balance}} = |\log \|W_{\text{mlp}}\|_2 - \log \|W_{\text{attn}}\|_2|$ to the training objective.
3. **Architecture search:** Penalize configurations with $\rho < 0.5$ or $\rho > 2$.

6.2 Implications for Pruning

Models with balanced pathways maintain high effective rank, meaning their residual streams have many active subspaces. This distributed geometric redundancy suggests that such models may be more amenable to structured pruning (removing heads, MLP neurons), though we do not test pruning directly. In contrast, collapsed models concentrate their “mass” in a single direction, potentially leaving insufficient redundancy for structured pruning.

6.3 Implications for Training Stability

The 1:1 Rule may explain why certain architectures train more stably than others. If the residual stream collapses to rank-1 during training, gradient signals become aligned, causing optimization difficulties. Maintaining $\rho \approx 1$ preserves subspace diversity, allowing gradients to flow through multiple independent channels.

7 Conclusion

We have discovered a simple and predictive regularity governing transformer residual stream geometry: the **1:1 Rule**. When the spectral norms of the MLP and attention pathways are balanced ($\rho \approx 1$), the residual stream maintains high-dimensional geometric structure across all layers. When they are imbalanced, the stream collapses to rank-1.

This regularity predicts collapse across the tested range of model scales (270M–2.7B), four model families, and both tested activation functions. The threshold correctly predicted out-of-sample collapse in Pythia-1.4B, a model family and architecture absent from our initial discovery set, suggesting it is a *predictive* property of the transformer architecture.

Limitations

Linearization. Our analysis linearizes the residual update (eq. (2)), ignoring non-linearities and layer normalization. While this approximation is standard in dynamical systems analysis of deep networks, future work could incorporate the full non-linear dynamics.

Attention structure. We model attention via its output projection W_{attn} . The synthetic experiment (§5.5) mirrors the empirically observed delay in attention-dominated collapse, suggesting the linearized output projection captures the leading-order dynamical effect. Incorporating the full Q/K/V and softmax Jacobian remains future work.

Causality. Our evidence is observational (correlational). To establish causality, one could train models with a spectral balance regularizer and measure whether this prevents collapse.

Scale. All models analyzed are $< 3\text{B}$ parameters. It remains to be seen whether the 1:1 Rule holds at 7B, 70B, or larger scales.

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A Detailed Methodology

A.1 Weight Extraction

For each model, we load the pretrained weights from HuggingFace and extract:

- **MLP weights:** For SwiGLU, W_{up} and W_{down} from the gate-up fused projection (splitting the fused matrix into gate and up components). For GELU, W_{up} and W_{down} from the standard FFN.
- **Attention weights:** The output projection W_o from each attention block. For GQA, we compute $W_{\text{attn}} = W_o W_o^\top$ to obtain a $[d, d]$ matrix for the covariance propagation.

A.2 Lyapunov Iteration

Starting with $C_0 = \mathbf{I}_d$, we iterate:

$$C_{\ell+1} = \frac{A_\ell C_\ell A_\ell^\top}{\|A_\ell C_\ell A_\ell^\top\|_2},$$

where $A_\ell = \mathbf{I} + W_{\text{mlp}}^{(\ell)} + W_{\text{attn}}^{(\ell)}$. The spectral norm normalization preserves the condition number and effective rank while preventing overflow.

A.3 Computational Details

All computations use double-precision floating point (float64) for the covariance matrices to handle the extreme condition numbers (10^{12}). Eigenvalue decompositions use `torch.linalg.eigvalsh` with adaptive thresholds for numerical zero detection.

A.4 Code Availability

The full analysis code, including the Lyapunov analyzer, architecture inspector, and plotting utilities, is available at <https://anonymous.4open.science/r/the-1-1-rule-FD86>.

B Sensitivity Analysis

B.1 Normalization Strategy

We tested three normalization strategies for the Lyapunov iteration: trace normalization, spectral norm normalization, and Frobenius norm normalization. All three yield qualitatively identical results (same ranking of models by stability, same collapse layers), confirming that the 1:1 Rule is robust to the choice of normalization.

B.2 Attention Approximation

We compared two ways of incorporating attention:

1. $W_{\text{attn}} = W_o$ (direct output projection, valid for MHA)
2. $W_{\text{attn}} = W_o W_o^\top$ (covariance contribution, valid for GQA)

Both approximations yield the same qualitative conclusion: Qwen and Llama are stable; all others collapse.

B.3 Excluding Attention

When attention is excluded from the Lyapunov iteration ($A = \mathbf{I} + W_{\text{mlp}}$), all models collapse immediately (layer 1–2). This confirms that attention plays a *critical stabilizing role* in the residual stream geometry.

C Independent-Basis Control

We repeat the synthetic experiment (§5.5) with W_{mlp} and W_{attn} generated from *independent* random orthogonal bases (no shared singular vectors, no anti-alignment). The bifurcation inverts: extreme ρ yields high effective rank, while balanced ρ collapses. For example, with $d = 512$, 5 seeds, and spectral normalization:

- $\rho = 0.10$: min erank = 48.15
- $\rho = 0.97$: min erank = 14.44
- $\rho = 20.00$: min erank = 90.66

This confirms the 1:1 Rule requires correlated, not merely random, structure between pathways.